Fiscal Year:	FY 2014	**Task Last Updated:**	FY 05/19/2014
PI Name:	Elefteriou, Florent Ph.D.		
Project Title:	Vestibular Stimulation: A Countermeasure For Bone Loss During Long-Term Space Mission?		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology (2) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	37232-0575	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
Start Date:	08/01/2012	**End Date:**	07/31/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Besnard, Stephane (INSERM, University of Caen Basse-Normandie (FRANCE)) Denise, Pierre (Université de Caen Basse-Normandie) Makishima, Tomoko (University of Texas, Galveston) Perrien, Daniel (Vanderbilt University)		
Grant/Contract No.:	NNX12AL35G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Background: We have demonstrated that loss of vestibular information following bilateral lesions in rats and mice decreases lower limb bone density, and that this effect may be mediated by activation of the sympathetic nervous system (SNS). Based on the knowledge that reduced vestibular information, increased sympathetic activation, and bone loss are common hallmarks of long-term space missions, we hypothesized that space-induced bone loss has a vestibular and sympathetic origin, and that vestibular stimulation might increase bone density. The main objective of this proposal is to evaluate the effect of vestibular stimulation on bone mass, with the goal of designing a new therapeutic approach to prevent bone loss in space. The project is divided into two specific aims. The first one will define the short and long-term effects of vestibular hair cell dysfunction on the vegetative system and on bone mass in mice, in order to determine whether this effect is compensated with time. Catecholamine levels, brown adipose tissue Ucp1/Gapdh expression, 3D-microcomputed tomography, and quantitative static and dynamic bone histomorphometry will be used. The second specific aim will characterize the effects of vestibular stimulation on the vegetative system and bone. Different centrifuge protocols (continuous and discontinuous, 1.5G/2G/2.5G) and galvanic vestibular stimulation will be used. Centrifugations will be performed in Dr. Makishma (UTMB, USA) and in Dr. Besnard and Prof. Denise (University of Caen, France) laboratories. Galvanic stimulation will be performed in Dr. Elefteriou's laboratory, in collaboration with Dr. Wood (Houston/UTMB, USA). All the bone analyses will be performed at Vanderbilt University (Dr. Elefteriou, Vanderbilt University). The part of the study performed in France is supported by the University of Caen Basse-Normandie and by the CNES. The ultimate goal of this multi-disciplinary effort is to determine whether vestibular stimulation could have beneficial effects on bone mineral density and sympathetic function during extended space mission. These findings may have application on Earth as well for the treatment or prevention of low bone mass conditions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research seeks to understand if and how structural and functional alterations in the vestibular system lead to changes in bone remodeling. This work is expected to impact our understanding of microgravity-induced bone loss, which is relevant to extended travel duration of future space missions, but also to fracture risk in patients with vestibular disorders on Earth.
Task Progress:	Aim1. To define the short and long-term effects of vestibular loss on the vegetative system and on bone mass in mice. Vestibular lesions (VBX) have been performed in all proposed groups for bone analyses 1, 3, and 6 months post-lesion. Group A (1 month VBX) has been analyzed by uCT and histomorphometry. Significant bone loss and architectural changes were observed in the distal femoral epiphyses of VBX mice compared to sham-operated mice (both genders). Histomorphometry analyses revealed a significant reduction in the bone surfaces covered by osteoblasts in VBX mice, and increase in osteoclast number. These results validate the effect of arsenylate lesions on bone in mice, and serves as our main reference to assess the effect of time on VBX-induced bone loss. The second and third group of mice (3 and 6 months post lesion) have been analyzed as well. uCT analyses did not reveal significant bone changes. These results indicate that the effect of vestibular lesions on bone is transient and compensated for with time. We also collected brown adipose tissues to assess Ucp1 expression as readout of sympathetic outflow. Ucp1 expression was found upregulated in the VBX group 1 month post-lesion but not 3 and 6 months post-lesion, consistently with the bone data. These results suggest that peripheral sympathetic outflow is increased in response to VBX and contributes to VBX-induced bone loss. Lastly, we collected bone marrow stromal cells (BMSCs) from sham and VBX mice to assess a possible effect of VBX on osteoblast progenitor differentiation or proliferation. Cfu-ap (=osteoblast commitment) and cfu-ob (osteoblast differentiation) assays were performed using BMSCs collected 1 month post-lesion. No significant changes were detected in both assays. These negative results suggest that VBX induces bone loss likely by reducing osteoblast proliferation, differentiation, or function in vivo rather than by reducing commitment to the osteoblasts lineage (in that latter case a reduction of cfu-ap colonies would have been observed). These results are in line with previous results and with a contribution of the sympathetic nervous system in VBX-induced bone loss. Aim 2. To characterize the effects of vestibular stimulation on the vegetative system and bone. a) Centrifugations. We have met with Dr. T. Makishima at the 2013 NSBRI meeting in Galveston to discuss the details of the centrifugation experiments, and also kept regular contact with Dr. Besnard in France to coordinate our efforts. The centrifuge (with 2 arms, hence double capacity) in Galveston is still not operational, but is expected to be early summer 2014. The centrifuge in Dr. Besnard laboratory is now operational and pilot experiments using WT and Het-/- mice are ongoing. b) Galvanic vestibular stimulation. The first phase of this experiment, which consists in establishing the Hindlimb Suspension (HS) model, has been completed, in collaboration with Dr. D. Perrien at Vanderbilt. IACUC protocol, custom cages, and mouse care are now in place, and our pilot experiment confirmed proper animal care and unloading-induced bone loss with our new setting. We have met Dr. Wood and Dr. Makishima in Galveston to discuss the details of the GVS stimulation experiment. The stimulator is now in our possession and the occurrence of nystagmus with this apparatus as been confirmed by Dr. Makishima, using eye tracker recording system in place in Galveston. WT mice are currently tested for their response to daily GVS stimulation, in absence of hindlimb suspension (HS). Depending on the results of this experiment, the second phase of the proposed experiment will be performed (GVC+HS).
Bibliography Type:	Description: (Last Updated: 08/16/2018)
Abstracts for Journals and Proceedings	Vignaux G, Elefterious F. (reported as G. Vignaux, S. Besnard, B. Philoxène, J Ndong, P. Mulder, P. Denise, F. Elefteriou) "Vestibular mass regulation via the sympathetic nervous system. (reported as Vestibular CNS afferences contribute to the regulation of bone mass.)." 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. , Feb-2013

Articles in Peer-reviewed Journals	Vignaux G, Besnard S, Ndong J, Philoxène B, Denise P, Elefteriou F. "Bone remodeling is regulated by inner ear vestibular signals." J Bone Miner Res. 2013 Oct;28(10):2136-44. http://dx.doi.org/10.1002/jbmr.1940 ; PubMed PMID: 23553797 , Oct-2013
Articles in Peer-reviewed Journals	Vignaux G, Ndong J, Elefteriou F. "Inner ear vestibular signals regulate bone remodeling through the sympathetic nervous system." PNAS (under review, as of May 2014). , May-2014